1. Only 25% of the intensity of a polarized wave passes through a polarizing filter. What is the angle between the electric field and the polarization axis of the filter?

2. A 200 mW vertically polarized laser beam passes through a polarizing filter whose axis is 35° from horizontal. What is the power of the laser beam as it emerges from the filter?

3. Unpolarized light of intensity I_0 is incident on two polarizing filters. The transmitted light intensity is $I_0/10$. What is the angle between the axes of the two filters?

4. Unpolarized light of intensity I_0 is incident on a stack of 7 polarizing filters, each with its axis rotated by 15° cw with respect to the previous filter. What light intensity emerges from the last filter?

5. At one instant, the electric field at one point of an electromagnetic wave are:

$$\vec{E} = 200 \sin\left(1 \times 10^7 x - 2 \times 10^{15} t\right)$$

   A. What is the velocity of propagation of the wave?

   B. What is the index of refraction of the material?

   C. What is the magnitude of The poynthing vector?

6. The intensity of sunlight on the Earth is I = 1360 W/m^2

   A. What is the power output of the sun?

   B. What is the intensiy of sunlight reaching Mars?

7. Use **GeoGebra** to:

   A. Define a functions that takes in the Incident intensity and polarization angle, and outputs the transmitted intensity

   B. Then use a **Slider** to produce a plot of what would happen to solar light incident on a polarizer, as the polaization angle swung through a full 360°

GeoGebra hints

Type each line into the Input Bar. A function can take *two* inputs — here `a` is the incident intensity and `t` is the angle in degrees. Do **not** name the function `I` — GeoGebra reserves `i`.

```
Tr(a, t) = a cos(t°)^2      Tr(1360, 60)      Tr(1360, x)
ang = Slider(0, 360, 1)     P = (ang, Tr(1360, ang))
```

Type the degree sign with **Alt+O**, or pick it from the on-screen keyboard. No degree sign handy? `a cos(t pi/180)^2` is the same thing. `Tr(1360, 60)` is a single number, a check that your function works. `Tr(1360, x)` draws the whole curve: right-click the Graphics view → **Graphics...** and set x from 0 to 360, y from 0 to 1500. Drag the slider and watch `P` ride along the curve. Before you trust the plot, ask yourself: *is sunlight polarized?*